

Question Number	Answer	Mark
4(a)(i)	The only correct answer is C 1250 minutes <i>A is not correct because it would take 75 000 seconds or 1250 minutes</i> <i>B is not correct because it would take 75 000 seconds or 1250 minutes</i> <i>D is not correct because it would take 75 000 seconds or 1250 minutes</i>	(1)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	(measure the) {image size / length of pollen tube} and then divide by {magnification / 100}	accept correct manipulated equation but ignore unmanipulated equation accept $A = I/M$ ignore magnitude accept calibrate an eyepiece graticule with stage micrometer (to calculate the actual size) ignore use graticule unqualified	(1)

Question Number	Answer	Additional guidance	Mark
4(b)	An explanation that makes reference to four of the following points: - the female gamete(s) and the male nuclei are haploid (1) - female gamete nucleus was fertilised by {one / a} male {nucleus / gamete} (to form the embryo cell) (1) - embryo (cell) is {2n / diploid} (1) (whereas two) polar nuclei were fertilised by {one / a} male {nucleus / gamete} (to form the endosperm cell) (1) - an endosperm (cell) is {3n / triploid} (1)	accept {egg cell/ovum} for female gamete can refer to tissue or cell ignore sperm reject generative nucleus reject generative nucleus accept zygote is {2n / diploid} ignore descriptions of diploid reject generative nucleus	(4)

Question Number	Answer	Additional guidance	Mark
4(c)	An answer that makes reference to five of the following points: • (drug containing <i>L. candidum</i>) extract tested on human (skin) {cells / tissues} (1) • appropriate {dosage / concentrations} (of drug containing <i>L. candidum</i> extract) identified / identification of {side effects / toxicity} (1) • (phase 1 / preliminary) tests on (people / volunteers) who {are healthy / do not have skin burns} (1) • (phase 2) - drug tested on {small / 100 to 500} groups of (patients / people) who have (skin) burns (1) • (phase 3) - drug tested on {large groups of / 1000 to 3000} (patients / people) who have (skin) burns (1) • double blind test involves one group receiving extract and the other receiving current (skin burn / hospital) treatment (1)	accept tested on animals ignore 'safe' / safety of drug ignore patients unqualified ignore patients unqualified ignore double blind unqualified / placebo group accept compare extract data to current skin treatment ignore peer review	(5)